Physics potential : **Top quark physics**

LHeC@1.2 TeV vs. LHμC@3.7 (5.3) TeV

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Outline: LHeC@1.2 TeV vs. LHμC@3.7 (5.3) TeV

- Single top-quark production at LHlC@LO
- Top-quark pair production at LHlC@LO
- Comments on inclusive cross-sections at the NLO accuracy

Single top-quark production at LHeC/LHμC

- Hard process (charged-current (CC) single-top) : At lepton-proton colliders, single-top production is dominated by CC DIS, mediated by a t-channel W exchange (5F scheme, b as a parton).
- LO & NNPDF proton PDFs
- Inclusive cross-sections calculations (**NO** GEN-level cuts)

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p e- > t~ vl
```

$$e^- p \to \bar{t}\,\nu_e + X$$

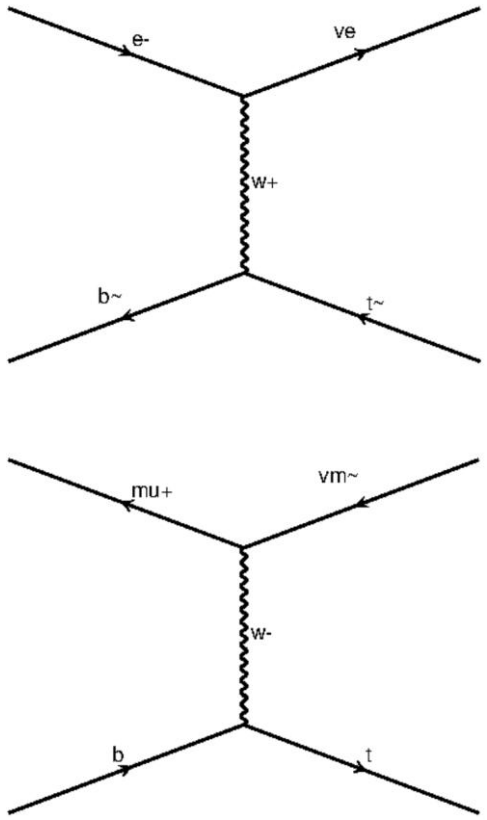
LHeC@1.2 TeV= $\sigma_{\rm GEN} = 1.6067304$ pb

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p mu+ > t vl~
```

$$\mu^+ p \to t\,\bar{\nu}_\mu + X$$

LHμC@3.7 TeV = $\sigma_{\rm GEN} = 21.242199$ pb

LHμC@5.3 TeV = $\sigma_{\rm GEN}$ = 36.7826 pb

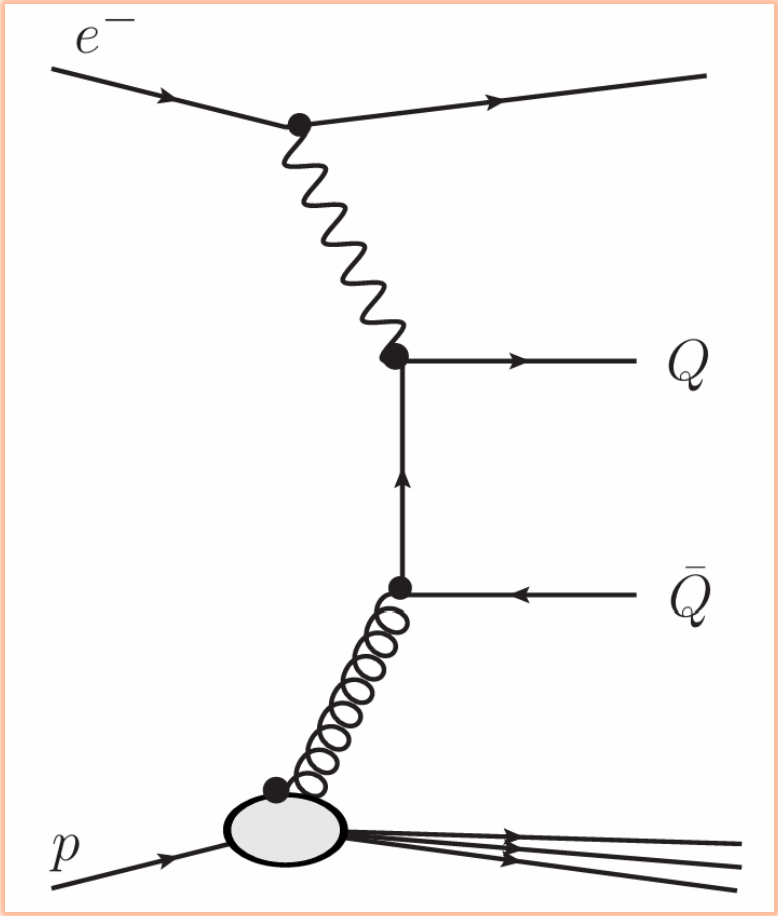


LO 2→2 diagrams in the 5-flavor scheme (5FS)

Top-quark pair production at a lepton-hadron collider

- Dominant LO mechanism: direct photoproduction via boson-gluon fusion $\gamma g \to t\bar{t}$; (photon radiated from the lepton, gluon from the proton).

- Sensitive to the gluon PDF



Direct photoproduction

Factorized rate $\sigma(lp \to l + t\bar{t} + X)$; $l = e, \mu$

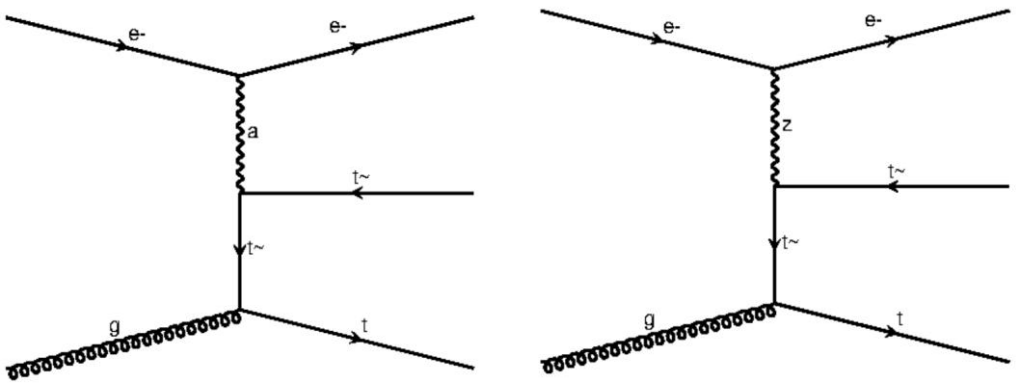
$$\sigma_{eh\to X} = \sum_j \int dx_\gamma dx_j f_\gamma^e(x_\gamma, \mathbf{Q}^2_{\max}) f_j^h(x_j, \mu_F, \mathtt{LHAID}) \hat{\sigma}_{\gamma j \to X}(x_\gamma, x_j, \mu_F, \mu_R)$$

top-quark pair production at LHeC/LHμC

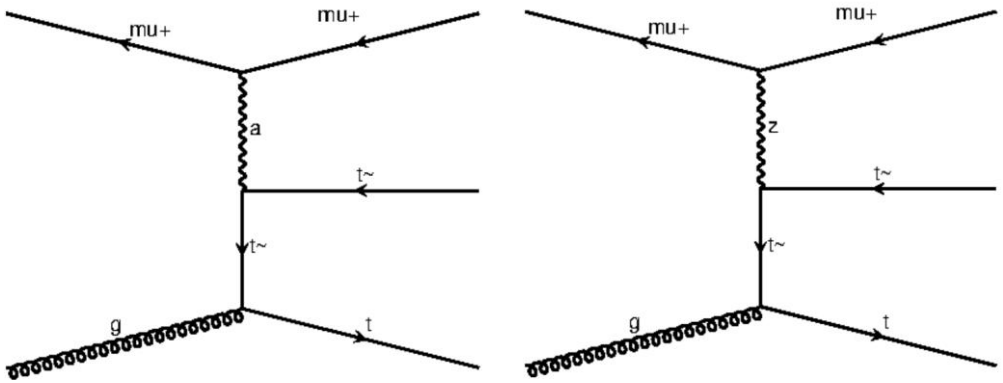
- At lepton-proton colliders, top-pair production is dominated by neutral-current (NC) interactions: the lepton emits a virtual photon (and also Z), which interacts with a gluon from the proton to produce the pair. [DIS-like: large Q^2 & Photoproduction-like: small Q^2]

- Underlying hard subprocess (dominant): $\gamma^* g \to t\bar{t}$ (and $Z^* g \to t\bar{t}$ is subleading)

$p\, e^- \to e^-\, t\bar{t}$



$p\, \mu^+ \to \mu^+\, t\bar{t}$



top-quark pair production at LHeC/LHμC

$$p\,e^- \to e^-\,t\bar{t}$$

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p e- > t t~ e-
```

$$p\,\mu^+ \to \mu^+\,t\bar{t}$$

```
import model sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
generate p mu+ > t t~ mu+
```

- LO & NNPDF proton PDFs
- GEN-level cuts applied (run card): $p_T^\ell > 0.1\ \mathrm{GeV}, \qquad |\eta_\ell| < 9$

LHeC@1.2 TeV = $\sigma_{\mathrm{GEN}} = 0.018435\ \mathrm{pb}$

LHμC@3.7 TeV = $\sigma_{\mathrm{GEN}} = 0.96053\ \mathrm{pb}$

- Inclusive cross-sections calculations (**NO** GEN-level cuts)

LHeC@1.2 TeV = σ_{GEN} = 0.03119 pb

LHμC@3.7 TeV = σ_{GEN} = 1.0302 pb

LHμC@5.3 TeV = σ_{GEN} = 2.2299 pb

LO inclusive top-quark productions

Process	20 GeV LHeC	50 GeV LHeC	500 GeV LHμC	1 TeV LHμC
$\sigma(\ell p \to \nu_\ell\, t + X)$ [pb]	0.55241	2.89176	21.26473	36.77466
$\sigma(\ell p \to \ell\, t\bar{t} + X)$ [pb]	0.00146	0.03231	1.03040	2.23280

1) Inclusive LO cross sections for single-top and top-pair production at the 20 and 50 GeV LHeC and at the 500 GeV and 1 TeV LHμC.
2) For the LHeC cases an 80% electron beam polarisation is assumed.
3) Core LO processes (no explicit extra jet at ME level).
4) No generation cuts (ptl/etal/ptj/etaj off).
5) Same flavour scheme (5FS).
6) Fixed-order parton-level totals (i.e. no dependence on shower/ISR settings for the quoted σ).
7) LO matrix elements + NLO PDFs (MG default proton PDFs: lhaid = 230000 : NNPDF23_nlo_as_0119)

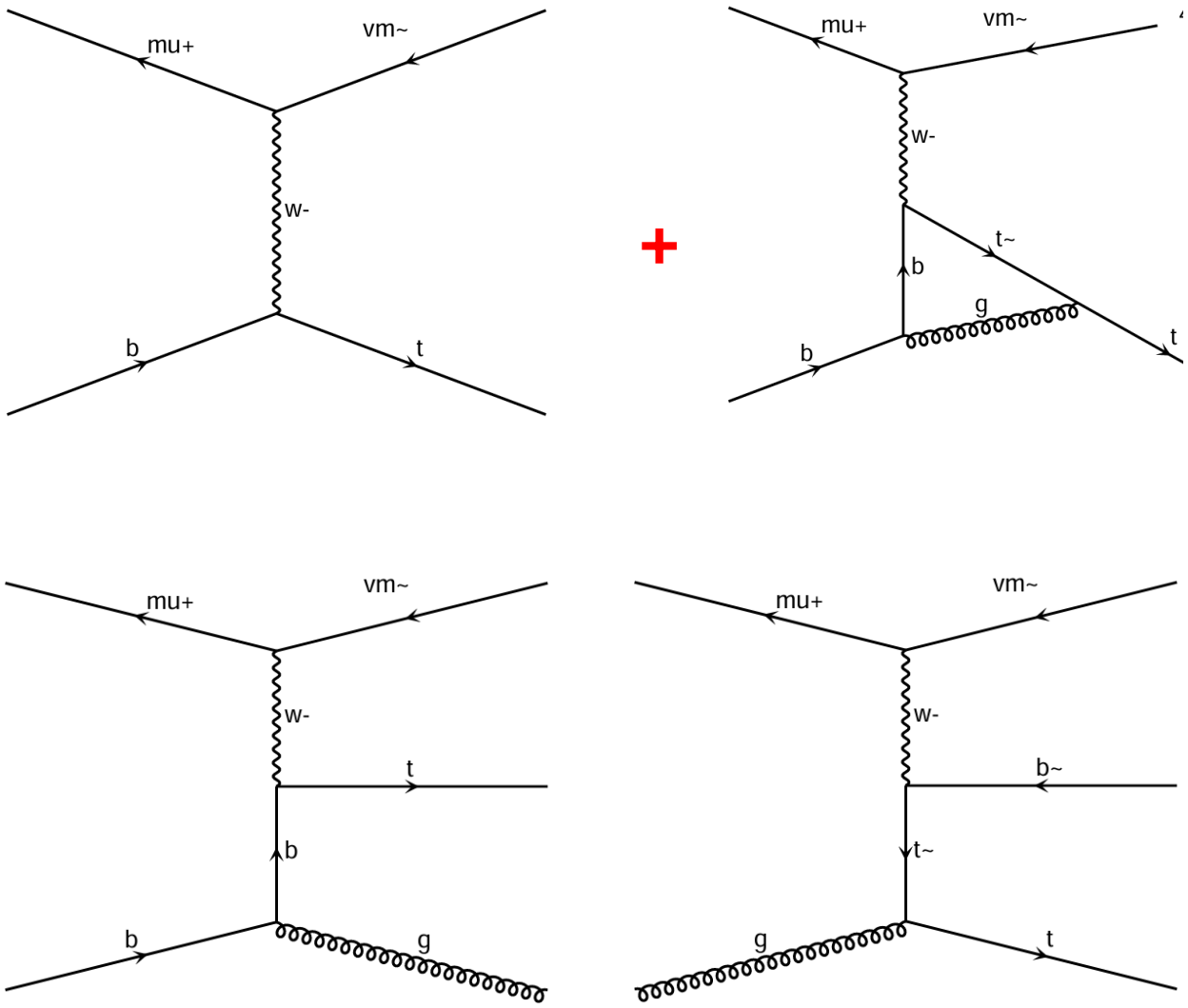
Inclusive cross-sections at the NLO accuracy

- CC NLO (single-top):

```
import model loop_sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~
define vl~ = ve~ vm~ vt~

generate p mu+ > t vl~ [QCD]
```

We import **loop_sm-full** to enable loop amplitudes, and the **[QCD]** tag instructs MG to compute the NLO QCD corrections.



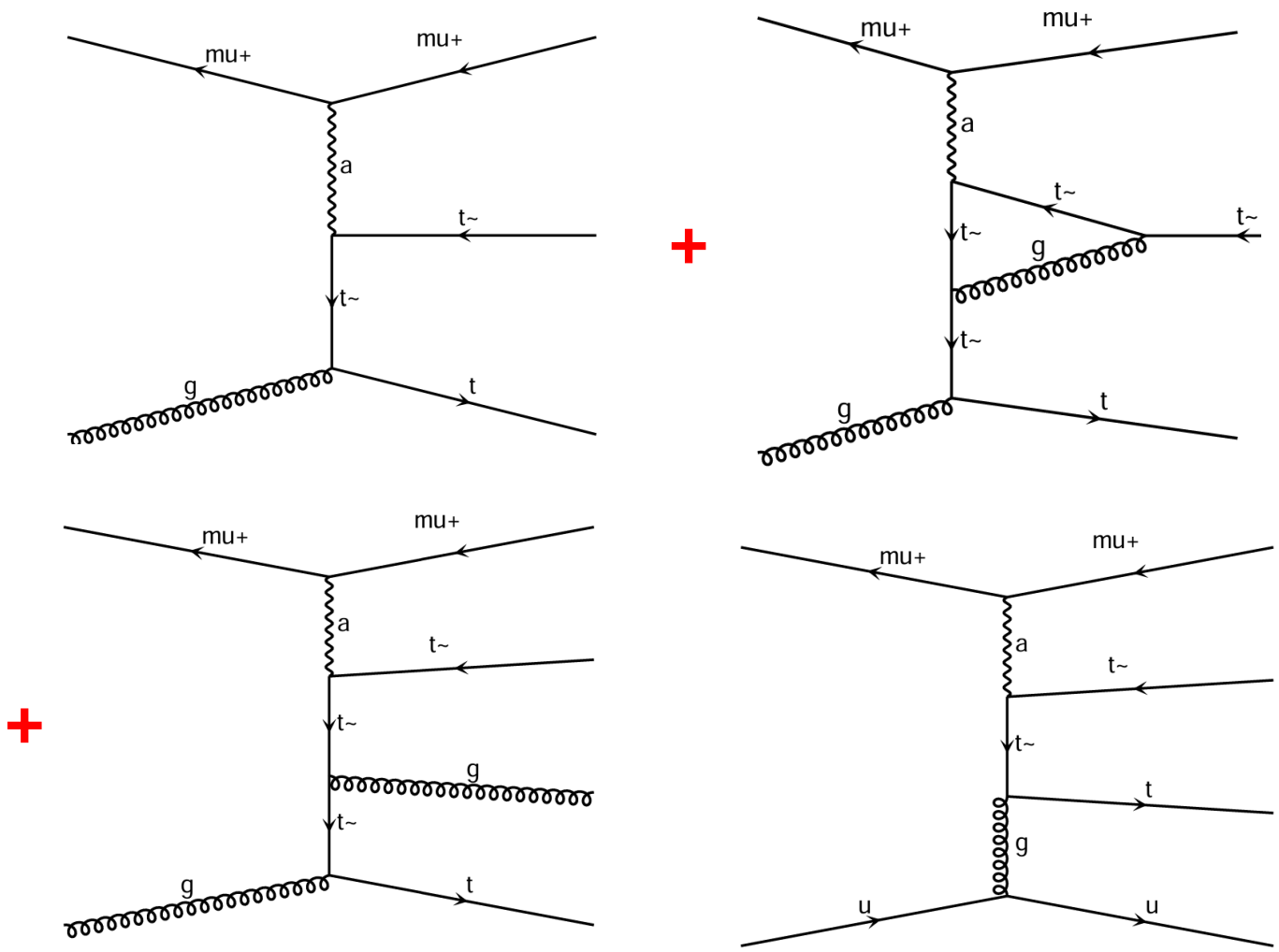
NLO QCD inclusive = $\sigma_{\text{NLO}} = \sigma_{\text{Born}} + \sigma_{\text{virtual(1-loop)}} + \sigma_{\text{real}(+1\text{ parton})}$

Inclusive cross-sections at the NLO accuracy

○ NC NLO ($t\bar{t}$):

```
import model loop_sm-full
define p = g u d s c b u~ d~ s~ c~ b~
define j = g u d s c b u~ d~ s~ c~ b~

generate p mu+ > t t~ mu+ [QCD]
```



Full NLO QCD corrections for the CC single-top and NC top-pair productions in *lp* collisions cannot be generated with public MG5_aMC@NLO due to the lack of DIS support at NLO!

$$\sigma_{\text{NLO}} = \sigma_{\text{Born}} + \sigma_{\text{virtual(1-loop)}} + \sigma_{\text{real}(+1\text{ parton})}$$

Sherpa3 for the NLO calculation [2503.17727 [hep-ex]]; no additional details are provided (generation cuts, ...)!

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Thank
you!